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ROLL NO : 25549

SECTION : CB

SUBJECT :Software Construction and Development

LAB NO : 03

TASK NO : 1

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main.cpp | Run ▶ | Output

```cpp

#include <iostream>
using namespace std;

int main() {
    cout << "Student Information" << endl;
    cout << "-------------------" << endl;
    cout << "Name: WAQAS SAMI" << endl;
    cout << "Age: 24" << endl;
    cout << "GPA: 2.9" << endl;

    return 0;
}

```

```
Student Information
-------------------
Name: WAQAS SAMI
Age: 24
GPA: 2.9
```

TASK NO : 2

C++ Online Compiler

main.cpp | Run ▶ | Output

```cpp

#include <iostream>
using namespace std;

int main() {
    // Display student information
    cout << "Student Information" << endl;
    cout << "-------------------" << endl;

    /*
       Declare variables to store
       student name, age, and GPA.
    */
    string name = "WAQAS SAMI";
    int age = 24;
    float gpa = 2.9;

    // Display the values of variables
    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "GPA: " << gpa << endl;

    return 0;
}

```

```
Student Information
-------------------
Name: WAQAS SAMI
Age: 24
GPA: 2.9
```

1. **What is a variable?**

A variable is a named memory location used to store data. For example, int age = 20; stores the value 20 in the variable age.

2. **What is cin?**

cin is used to take input from the user through the keyboard.

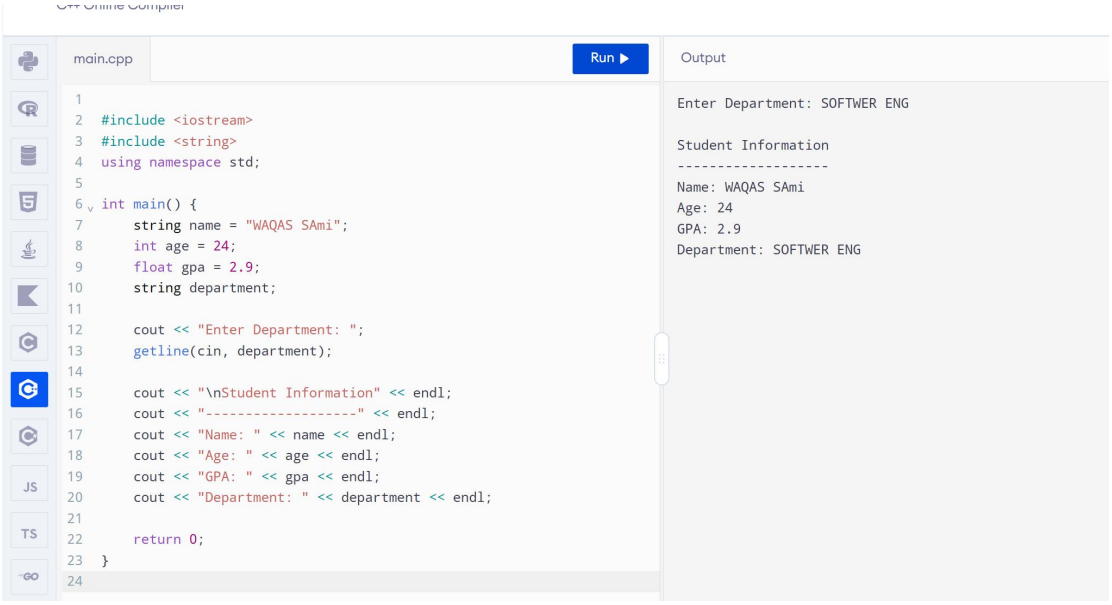
Example: cin >> age;

3. **What is cout?**

cout is used to display output on the screen.

Example: cout << "Hello";

TASK NO : 3



TASK NO :4

C++ Online Compiler

main.cpp

Run ▶

```
12      int birthYear;
13
14      // Take input from the user
15      cout << "Enter Age: ";
16      cin >> age;
17      cin.ignore();
18
19      cout << "Enter Department: ";
20      getline(cin, department);
21
22      // Calculate birth year
23      birthYear = 2026 - age;
24
25      /*
26         Display student information
27         and calculated birth year.
28      */
29      cout << "\nStudent Information" << endl;
30      cout << "-------------------" << endl;
31      cout << "Name: " << name << endl;
32      cout << "Age: " << age << endl;
33      cout << "GPA: " << gpa << endl;
34      cout << "Department: " << department << endl;
35      cout << "Birth Year: " << birthYear << endl;
36
```

Output

```
Enter Age: 24
Enter Department: SOFTWER ENG

Student Information
-------------------
Name: WAQAS SAMI
Age: 24
GPA: 2.9
Department: SOFTWER ENG
Birth Year: 2002
```

TASK NO :5

main.cpp

Run ▶

```
1
2   #include <iostream>
3   #include <string>
4   using namespace std;
5
6   int main() {
7       // Declare student variables
8       string name = "WAQAS SAMI";
9       int age;
10      float gpa = 2.9;
11      string department;
12      int birthYear;
13      char grade = 'A';
14      double fee = 25000.75;
15
16      // Take input from the user
17      cout << "Enter Age: ";
18      cin >> age;
19      cin.ignore();
20
21      cout << "Enter Department: ";
22      getline(cin, department);
23
24      // Calculate birth year
25      birthYear = 2026 - age;
26
```

Output

```
Enter Age: 24]
Enter Department:
Student Information
-------------------
Name: WAQAS SAMI
Age: 24
GPA: 2.9
Department:
Birth Year: 2002
Grade: A
Fee: 25000.8
```